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Maximizing neutron yields by scaling hollow diameter of a dense plasma focus anode

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Experiments were performed to maximize the neutron yield from a 2 kJ dense plasma focus (DPF) and characterize the amount of copper sputtered from the surface of an anode by varying the diameter of the anodes' on-axis hollow. The hollow is a void in the copper material along the longitudinal axis of the anode. All the anodes had an outer diameter of 1.2 in. and the diameter of the hollow varied from 0 in. (no hollow) to 1 in. The anodes with a hollow produced a greater number of neutrons per discharge than the anode without a hollow. Over 40 discharges, the hollow anode that yielded the most neutrons ($9.1 \pm 0.4 \times 10^6$ neutrons per discharge produced with the 0.75 in. hollow) produced >6 times more neutrons than the anode with no hollow. A qualitative observation of the anodes after 130 discharges showed less surface damage on anodes with a larger hollow. Quantitative sputter measurements were performed by characterizing the amount of copper sputtered onto on-axis quartz targets for three newly machined anodes, each with a particular hollow diameter. The quantitative results matched the qualitative observations: the copper sputter was reduced using larger hollows. The largest hollow sputtered 17 ± 1.0 nm/sr/discharge of copper, a reduction of 69% compared to the anode with the most damage. <https://doi.org/10.1063/1.5051665>

I. INTRODUCTION

A common goal of neutron generating devices is maximizing the output neutron yield for a given amount of stored energy. A dense plasma focus (DPF) is a compact fusion device for neutron production. Many design features of the DPF anode,^{1–5} cathode,⁶ and insulator^{7–9} have been shown to affect a device's neutron production. DPF operating parameters, such as charge voltage,^{1,5} peak current,¹ and background gas properties,^{10,11} have also been studied to maximize a DPF's neutron output. We investigate a correlation between a geometric property of the anode, other than the outer diameter or tapered surfaces, in order to increase neutron yields while keeping all other DPF parameters constant. We focused on the anode hollow diameter because it is an optimization parameter which did not change the timing of the DPF circuit, a measurable amount in our experiments.

A DPF is a pulsed power driven fusion device consisting of a coaxial anode and a cathode separated by an insulator inside a vacuum chamber filled with a low-pressure gas.^{12–14} A voltage source is applied in a pulsed manner and produces a plasma sheath along the surface of the insulator in the anode–cathode gap. The plasma sheath is accelerated by the $\mathbf{J} \times \mathbf{B}$ force down the length of the anode, ionizing and sweeping up neutral gas. After the sheath reaches the end of the anode, the sheath is accelerated radially inward and implodes on axis, creating a pinch, which can result a short burst of radiation. If using deuterium as a fill gas, neutrons can be produced. One theory for the neutron production mechanism is beam–target fusion, where the beam is created

by the electric fields^{15,16} that occur from an $m = 0$ instability. These electric fields will accelerate ions and electrons in opposite directions. Ions are accelerated away from the anode, and electrons are accelerated toward the face of the anode.

Damage to the surface of the anode is produced primarily by intense electron bombardment to the front surface of the anode.¹⁴ The material ablated by electron bombardment of the anode has been shown in past shadowgraphy experiments¹⁹ to begin emitting tens of nanoseconds after the pinch and can continue for more than a microsecond. This damage has been shown to be significantly reduced when the center electrode has negative polarity.^{17,18} A less severe mechanism of producing surface damage is attributed to plasma recoil¹² along the exposed anode surface. Both of the mechanisms which damage the anode surface can expel copper from the anode, which is referred to as “sputter” in this paper. The time scale of the surface damage produced by plasma recoil may sputter copper into the pinch-region, resulting in a contaminated pinch.

The reduction of the anode sputter to the DPF community is most relevant to DPF experiments which use solid targets placed inside the vacuum chamber close to the pinch to provide a solid physical target source for beam–target neutron production. Deuterated targets are primarily used with a deuterium fill gas and have been shown to increase neutron production.^{20–22} When using solid targets, the copper sputter will coat the targets, and as a copper surface layer thickness increases, the intended target material may be shielded from the DPF beam on subsequent discharges. Reducing the rate of copper sputter will increase the useful lifetime of a target.

The geometry of the anode will affect many of the transport properties of the plasma sheath. One anode property used previously on DPF devices is an on-axis hollow.^{23–26}

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Although previously published DPF experiments have used hollow anodes, there is no published systematic study of the effect of the hollow size on the neutron yield. This paper reports the experimental results of the neutron yield and the anode surface sputter when varying the diameter of a hollow.

II. EXPERIMENT

A. DPF device layout

The 2 kJ DPF used in these experiments has a circuit with a capacitance of $3.2 \mu\text{F}$, an inductance of $38 \pm 2 \text{ nH}$, and a charge voltage of 36 kV. A peak current of $250 \pm 11 \text{ kA}$ is discharged through the device, with a $500 \pm 18 \text{ ns}$ rise time. A fill pressure of 1.3 Torr deuterium (D_2) produces the correct timing for the sheath to pinch while the circuit reaches the peak current $\pm 25 \text{ ns}$. The reduction in current produced by the pinch is on average $-110 \pm 20 \text{ kA}$ for the 0.75 in. hollow anode. The uncertainty values are calculated as the standard deviation of the shot-to-shot fluctuations of the device. A plot of a typical discharge current trace, produced with a 36 kV charge voltage and a fill pressure of 1.3 Torr deuterium, is shown in Fig. 1.

A schematic of the DPF vacuum chamber and relative location of diagnostics discussed in this paper are shown in Figs. 2(a) and 2(b). A removable gate valve is located downstream of the DPF chamber to block the downstream diagnostics from the copper sputter. A blank gasket with a 1.0 in. aperture is located on-axis between the gate valve and the downstream vacuum chamber [labeled in Fig. 2(a) as the “Sputter Target’s Chamber”] which houses four quartz discs used to capture and quantify the sputter. The 1.0 in. aperture upstream of the targets shielded the unused targets from the copper.

The anodes used in this experiment have the same length, outer diameter, and radius of curvature on the outer edge [shown in Fig. 2(c)]. Each anode has a particular diameter of the hollow, from 0 in. (no hollow) to 1 in. The anodes were machined from oxygen-free, high conductivity (OFHC) copper. The portion of the anode exposed to the plasma sheath has a 1.2 in. outer diameter and a length of 1.02 in. The anode’s aspect ratio $A = L/2a$, where L is the effective anode length and a is the radius of the anode, can be used to characterize the device as a Mather-type¹³ ($A > 1$, typical values 5–10), Filippov-type¹² ($A < 1$, typical values are 0.2)

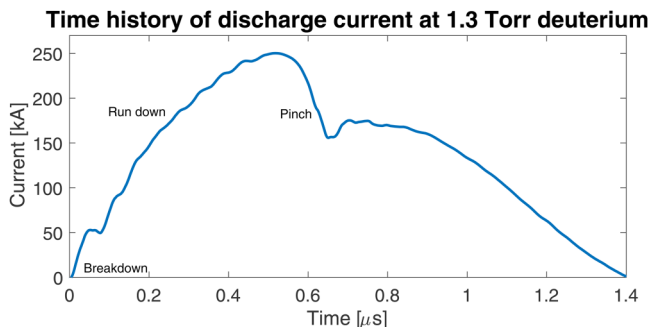


FIG. 1. A time history of discharge current for a typical DPF discharge at 1.3 Torr deuterium.

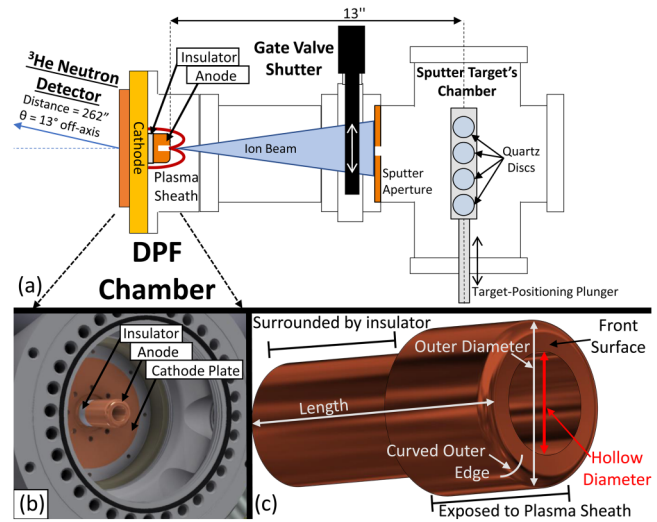


FIG. 2. (a) A schematic of the DPF chamber showing the location of the quartz targets used to capture sputter and the location of the ^3He neutron yield detector. (b) A CAD model of the DPF chamber. The anode extends inside the insulator, which separates the anode from the cathode plate. (c) A rendering of a hollow anode used in this study. The anode features that are discussed in Sec. II A are labeled.

device, or a hybrid Filippov-Mather ($A \sim 1$). The aspect ratio of the hybrid anodes used in this experiment is $A = 0.85$.

The front surface of the anode intersects the hollow, with a sharp edge joining the surfaces. The outer diameter and the front surface are joined with a rounded edge with a radius of curvature of 0.1 in. Each anode was discharged 40 times to study its neutron yield. Prior to these discharges, each of the anodes was new, with only 50 previous discharges for a separate experiment.

B. Neutron yield detector

A ^3He detector²⁷ is used to measure neutron yields from the DPF. The physical properties of the detector, its technique of producing counts from the neutron flux, and its calibration technique are described in detail.²⁵ ^3He detectors are not sensitive to the high-energy neutrons produced on a DPF, and thus blocks of polyethylene are used to thermalize the neutrons to a detectable energy. The ^3He detector’s calibration to convert counts to neutron yield ($9.8 \pm 0.9 \times 10^3$ neutrons/count) was performed using an isotropic 93 nC ^{252}Cf source inside the DPF chamber. The location of the ^3He detector used to measure the neutron yield is labeled on the left of Fig. 2(a). The position of the detector is optimized for good counting statistics with $<10\%$ uncertainty in calibration and nearly no signal pileup.

C. Copper sputter measurements

A qualitative trend was observed for the amount of surface damage produced on the anodes during the measurement of the mean neutron yield as a function of hollow diameter (discussed in Sec. III A). A follow-up measurement was performed to quantify the amount of copper sputtered from the anode surface. Three newly machined anodes with various hollow diameters were discharged 200 times and a

target-positioning plunger [shown in Fig. 2(a)] with four round 2 in. diameter quartz disks was placed on-axis 13 in. from the anode. By changing how far the target-positioning plunger is moved into the chamber, each target can be positioned on-axis while the other targets are shielded. Each target was exposed to the copper sputter from the DPF for 50 discharges. Three mass and thickness measurements were collected on each disc before and after the copper sputter was collected.

III. NEUTRON YIELDS

A. Effects of hollow diameter on neutron yields

To isolate the effects of the hollow diameters on DPF performance, the fill pressure and voltage of the DPF remained constant while the diameter of the hollow was varied. Changing the hollow diameter did not cause the rise time, peak current, and pinch time of the DPF to change beyond the natural shot-to-shot fluctuation of the device which are given in Sec. II A. Keeping these variables constant isolated the hollow diameter as the only variable changed during this study.

The yields produced during the hollow scan are shown in Fig. 3. All anodes with on-axis hollows produced higher average yields than the anode without a hollow. The average yield for the 0.75 in. hollow was >6 times larger than the average yield from the anode with no hollow. The error bars shown in Fig. 3 are the standard deviation (σ) of the yields measured for 40 DPF discharges, which are approximately 30%–45% of the mean. Standard deviations of this magnitude with respect to the mean have been reported for similar-scale DPF devices.^{28,29} The gray rectangles in Fig. 3 are the statistical uncertainty of the mean, calculated as the standard error of the mean (SEM). The results from a follow-up study to

measure the DPF's output distribution (discussed in Sec. III B) is shown in blue for comparison to the results for the 1.0 in. from the anode hollow scan. The yields measured for the seven different anode hollows and the distribution function study were performed months apart, indicating that the mean of neutron yield per discharge did not change over time.

Given the magnitude of the standard deviation of yields, for a single discharge, it cannot be stated which of the anodes with hollows ≥ 0.5 in. will produce the maximum yield. For data sets with many discharges, the average yield becomes less uncertain as the SEM decreases with increasing number of discharges. For 40 discharges, the 0.75 in. hollow produced the largest yield per discharge, with an uncertainty small enough to distinguish this as the highest average yield per discharge. The mean yield from the 0.75 in. hollow was $9.1 \pm 0.4 \times 10^6$ neutrons per discharge, and the 0.625 in. hollow anode produced the second-largest average yield of $8.2 \pm 0.4 \times 10^6$ neutrons per discharge.

B. Distribution function of DPF yields

To characterize the distribution of DPF neutron yield per discharge, an experiment was conducted using the 1.0 in. hollow anode by measuring the yield for 255 discharges while keeping all DPF parameters constant. The results, plotted in Fig. 4, show the measured neutron yield for a given discharge, a binned histogram of the neutrons per discharge, and a unimodal distribution of $f(x) = \alpha e^{[-(x-\mu)^2/(2\sigma^2)]}$ overlaying the binned data. The unimodal distribution function employs the parameters calculated from the raw data ($\mu = 6.1 \times 10^6$ neutrons/discharge and $\sigma = 2.6 \times 10^6$ neutrons/discharge). The unimodal distribution agrees with the histogram's data, indicating that the source of the standard

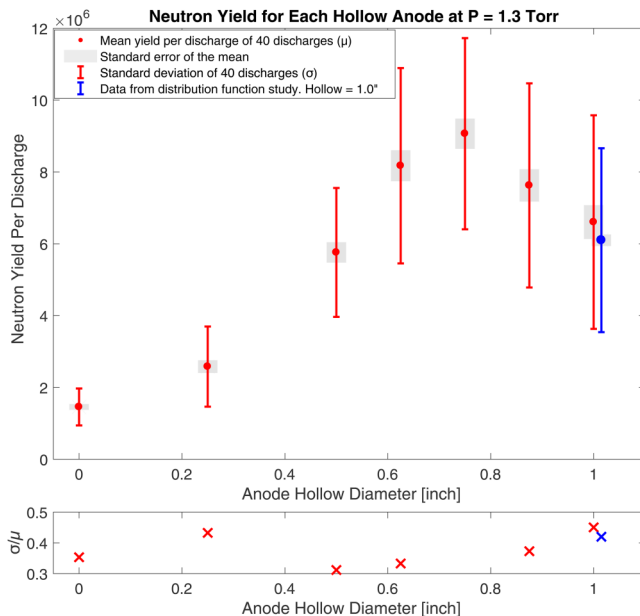


FIG. 3. Top: mean neutron yield per discharge (μ) as a function of the anode hollow diameter. Data points and error bars in red are the result of 40 discharges to study the hollow's effect on the neutron yield. The blue data point is the result of 255 discharges to study the DPF's distribution function of the neutron yield per discharge which is described in Sec. III B. Bottom: the ratio of the standard deviation to the mean yield per discharge.

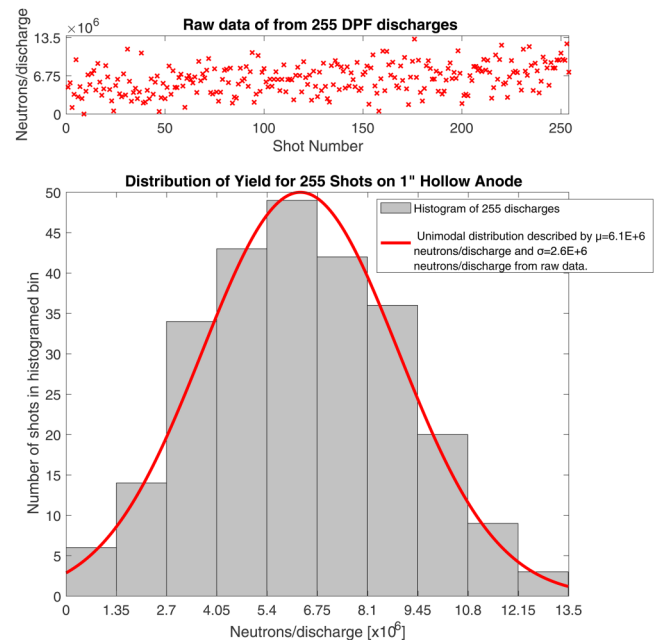


FIG. 4. (Top) raw data of neutrons per discharge. (Bottom) distribution function of 255 DPF discharges for identical experimental parameters. The overlaid red curve is a unimodal distribution function using the mean (μ) and standard deviation (σ) from the raw data.

deviation in the yields measured is the natural fluctuations of the DPF neutron-production mechanism.

IV. COPPER SPUTTER RESULTS

A. Qualitative observations

While performing the study of the neutron yield for each anode hollow diameter (see Sec. III A), different amounts of surface damage were observed on the anodes. Before the anode was installed, it appeared polished and undamaged. The macroscopic view of the newly machined anode [Fig. 5(a)] shows an unused, undamaged, polished copper finish. After being used for 130 DPF discharges, the surface of the same anode appears damaged [Fig. 5(b)]. The optical microscopic image of the unused anode [Fig. 5(c)] shows periodic score marks from the tool used to machine the anode. The microscopic view of the anode after surface damage is produced [Fig. 5(d)] no longer shows machining lines, and the surface appears to be suffering ablation damage.

A photograph of surface damage on the anodes after plasma interaction from 130 discharges is shown in Fig. 5(e).

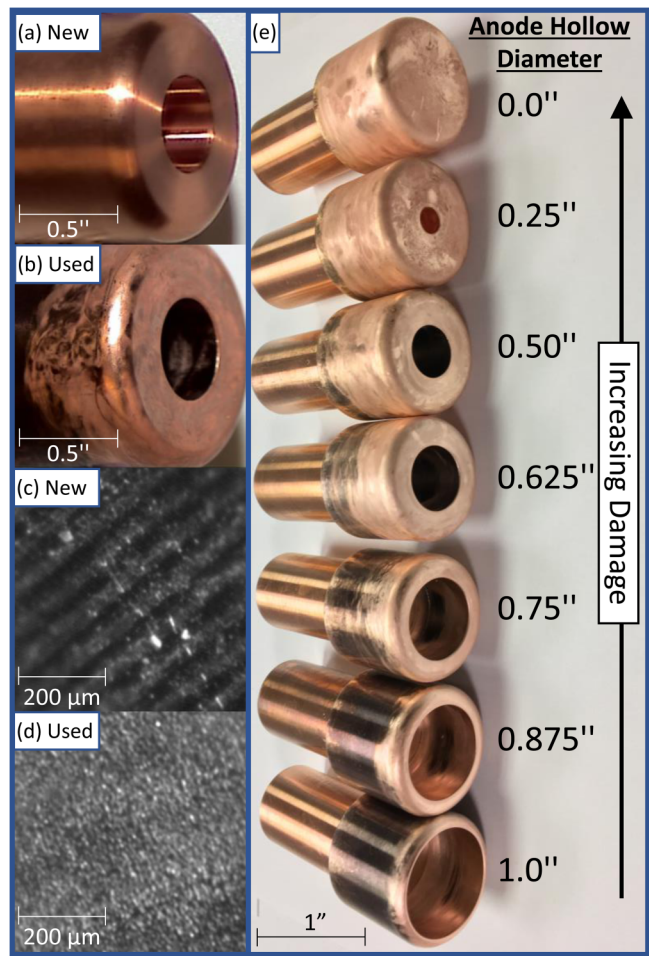


FIG. 5. (a) Photograph of a newly machined 0.5 in. hollow anode. (b) The 0.5 in. hollow anode after 130 DPF discharges. (c) An optical microscope view of the newly machined copper surface. (d) A microscope view of the copper surface after 130 DPF discharges. (e) Photograph of the set of anodes investigated in this manuscript after 130 discharges. A trend of more surface damage for smaller hollows is visible.

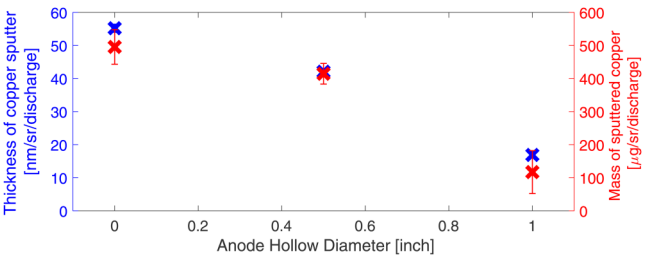


FIG. 6. Measured thickness (left y-axis) and mass (right y-axis) of copper sputtered on a series of on-axis quartz targets. Each data point is calculated from the amount of copper added after 200 DPF discharges over a solid angle of 1.85×10^{-2} sr.

The trend of larger anode hollows corresponding to less surface damage is visible. The mechanism producing the copper sputter is not fully understood; however, as surface damage increased, it appeared that copper was leaving the anode surface, because a copper layer began coating the inside walls of the vacuum chamber.

B. Copper sputter mass and thickness

The experiment to quantify copper sputtered from 200 DPF discharges is described in Sec. II C, and the results are shown in Fig. 6. The total copper added to the targets in 200 discharges is the sum of the values measured from each disk after 50 discharges. The error in the mass and thickness from 200 discharges is the standard deviation of the mass and thickness added to each disk.

The quantitative results match the qualitative trend observed in Fig. 5(a), as the copper sputtered from the anodes decreased with a larger hollow diameter. The largest hollows sputtered an average of 17 ± 1.0 nm/sr/discharge of copper and an added mass 117 ± 64.8 μg/sr/discharge. This is 76% less mass than the anode with no hollow and a reduction of thickness of 69%.

V. CONCLUSION

We report on optimizing DPF performance by varying the hollow diameter of the anode. During the study of the neutron yield as a function of hollow diameter, the largest average yield produced by an anode with a hollow ($9.1 \pm 0.4 \times 10^6$ neutrons per discharge) was >6 times larger than that of the anode with no hollow. Because the standard deviations were 30%–45% of the mean yields, it cannot be stated which of the anodes with hollows ≥ 0.5 in. will produce the maximum yield for a single discharge.

The distribution function of yield per discharge from 255 discharges with identical input parameters was a unimodal distribution described by the mean and standard deviation of the raw data. This supports that the magnitude of the standard deviation is caused by the natural shot-to-shot fluctuation of the device and that the SEM is a metric of the uncertainty of the mean yields per discharge. When averaging performance over many discharges, the uncertainty of the mean is less than the standard deviation, and the long-term performance of the 0.75 in. hollow produced a distinguishable maximum average yield over 40 discharges.

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Both qualitative and quantitative measurements of the copper sputtered from the anodes resulted in the larger hollows producing less sputter. The largest hollow sputtered a thickness of 16.9 ± 1.0 nm/sr/discharge of copper and an added mass of 117 ± 64.8 μ g/sr/discharge, which is 76% less mass and a reduction of thickness of 69% over the anode with no hollow. We conjecture that the relationship between the anode sputter produced by plasma recoil and the neutron yield production may be due to copper contamination in the pinch and bremsstrahlung radiation loss reducing both the pinch plasma temperature and the D-D fusion reaction cross-section. Although the time scale of the copper sputter from plasma recoil is not yet understood, if the copper expelled from the surface is present in the pinch, the concentration of deuterium in the pinch will decrease. When accelerated deuterons collide with copper, the beam loses energy and decreases the probability of fusion events. Follow-up experiments to measure a more complete data set of the copper sputter from hollows including the 0.625 in., 0.75 in., and 0.875 in. would allow for a stronger relationship between the copper sputter and the neutron yield.

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